ITSYSAD/LBYSYAD Case Study Specifications

Objectives

This case study is formulated for students to demonstrate the following:

- To analyze a given set of requirements from a use case
- To design, deploy, and manage a Linux-based server
- To demonstrate practical system administration skills
- To document the setup, configuration, and settings thoroughly

Background

Xcite Interactive is a game studio that used to develop mobile and online PC games, however due to their previous success, they have expanded to a medium sized company. In line with their expansion, they increased the size of their IT team in order to support many of their much-needed services and solutions.

The owners of Xcite have contacted your team once again, to design and deploy a server setup that would support their needs while being reasonably secure as well as only using free and opensource tools. The following are the design aspects needed from you and your team.

Firstly, the IT team should include the following users or roles:

A. System Admin (Head) – manages the different user accounts of the system, including the other admins and their tasks
B. Web Admin – manages the helpdesk web service of Xcite
C. Database Admin – manages the database used by the webpage and its backups
D. Network Admin – manages the network configuration, remote access privileges
E. Security Admin – manages the firewall and monitoring services

Secondly, the services needed would include the following:

A. Web Server – basic web-based help desksystem
   a. Install web service to host the helpdesk system (apache2)
   b. Install open-source helpdesk system UVdesk community helpdesk https://github.com/uvdesk/community-skeleton
   c. Configure and setup basic accounts
   d. Create sample requests to simulate usage of the helpdesk system
B. Remote Access
   a. Configure secure and encrypted remote access (ssh)
   b. Disable password-based authentication and implement SSH key-based authentication
C. Database Service
   a. Install and configure a simple database server (mysql-server) https://documentation.ubuntu.com/server/how-to/databases/installmysql/index.html
D. Monitoring Services – basic system monitoring measures
   a. Install monitoring tools (e.g. top, htop, vmstat, netstat, etc)
   b. Collect resource usage every 5 minutes
   c. Create a script that would return the resources used in a simplified format
E. Backup Service
   a. Create a script that would perform manual backup (bash and tar)

b. Perform daily backups of the database contents every 2AM to a separate backups folder containing appropriate filename
c. Perform archival of backups every after 1 week by transferring the backup files to a separate archive folder

F. Security Services – basic security hardening measures
a. Configure firewall to restrict access to necessary ports only (iptables or ufw)
b. Implement basic intrusion prevention (fail2ban)
c. Perform weekly system updates

Thirdly, to document the installation setup, configuration, and settings of the machine. The installation manual should include a very brief installation manual (max of 3-pages) to show the commands used to setup each of the services together with a brief description per command used. The document can be split into sections for each service containing a 2-column table format for each service, with the left column for the command and the right column for the brief description.

The configuration and settings manual can be split into sections for each service, containing configurations and/or settings used in the implementation. This should include the user credentials used among others.

Lastly, to record a 5-minute demonstration video (max of 8-minutes only) to present the different systems, their configurations and use, with subtitles. No need to include persons, only visual aids (such as presentation) as well as the server view (CLI and web-based GUI), to show a walkthrough of the system implementation as a whole. Focus on presenting your output in relation to the requirements given.

Grouping

For the groupings of the Case Study, use the same grouping in your LBYSYAD and upload the deliverables there. Make sure that all members are part of the same group.

Submission Deliverables

Document deliverables are to be uploaded to AnimoSpace (under the appropriate assignment) on before the indicated deadlines. Failure to upload any or all of the requirements would merit a score of 0 for each deliverable not submitted. ***Submission of deliverables outside AnimoSpace and beyond the deadline would not be accepted.***

1. Installation Manual (max of 3 pages)
2. Configuration and Settings Manual (max of 2-3 pages)
3. Encryption Keys (if any)
4. 5-minute Demonstration Video (max of 8-minutes and at most 720p resolution)

Any documentation in excess of the maximum number of pages would be truncated and not be included in your submission. Any demonstration video in excess of the number of minutes will also not be considered in the final output.